Diogo Estevão Pereira Pinto

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Education

University of Lisbon, PhD in Physics **2017 – 2022**

Thesis title:" Collective *dynamics of flexible active particles on substrates: from cells to tissues"*
Approved with Distinction and Honors

University of Lisbon, MSc in Physics **2015 – 2017**

Thesis title:" Adsorption *of colloidal particles on mobile rafts"*

University of Lisbon, BSc in Physics **2012 – 2015**

Current position

Assistant professor, University of Lisbon **Since 2025**

Researcher at the Centre for Theoretical and Computational Physics **Since 2025**

Research interests

My research focuses on the understanding of complex systems in soft matter and biophysics where collective behavior leads to emerging phenomena. Particularly, I have been successful in applying statistical physics tools and numerical modelling to study systems ranging from epithelial monolayers to nanoscale particles. The accomplishments presented in this CV, attest to the approaches and methods I developed through my scientific career and am now able to handle independently. As I move forward, I would like to develop novel and ambitious research that focuses on understanding the collective behavior of biological systems at different length scales. This focus can be subdivided into three main approaches:

- Understanding how the collective behavior of living systems leads to different rheological properties;
- Developing new tools and methods rooted in statistical physics and complex systems modelling to describe multi-cellular systems under deformations;
- Translating the research outcomes into new industrial technologies focused on disease control and regenerative medicine therapies.

Past research positions

UKRI Horizon Europe Guarantee MSCA Postdoctoral Fellow **2024 - 2025**

Under the supervision of Prof. Julia Yeomans, within the UKRI Horizon Europe Guarantee MSCA Postdoctoral Fellowship award reference EP/Z002761/1.
Research topics: Rheology and Fracture in Confluent Cell Layers.

Postdoctoral Researcher **2022 – 2024**

Under the supervision of Prof. John Russo, within the European Research Council Grant SOFTWATER at the University of Rome" La Sapienza".
Research topics: Inverse design methods for the self-assembly of complex structures.

Visiting research scholar **Aug – Dec 2018**

Under the supervision of Prof. M. Lisa Manning at the University of Syracuse, USA.
Research topics: Compartmentalization of epithelial monolayers through cell-substrate interactions.

Junior Research fellow **2015 – 2016**

Under the supervision of Prof. Nuno Araújo and Prof. Margarida Telo da Gama, with financial support from the Portuguese Foundation for Science and Technology (FCT) under Contract no. IF/00255/2013.
Research topics: Irreversible adsorption of DNA-functionalized colloidal particles on complementary mobile patches.

ERASMUS+ summer internship **Jul – Sept 2015**

Under the supervision of Prof. Eduardo Garcia at the Astroparticles and Nuclear Physics Group, University of Zaragosa, Spain.
Research topics: Underground experiments for the detection of dark matter.

Publications

1. *" Multi-phase field model reveals internal dissipation is crucial for spontaneous hole formation in cell monolayers"*, **Diogo E. P. Pinto**, Jan Rozman, and Julia M. Yeomans, Nat. Commun. **17**, 8288 (2026).
   Significance: Using a wet version of the multi-phase field model, we identified a new phase of collective behaviour where topological defects are able to spontaneously generate holes in a confluent cell monolayer and proposed relevant scaling laws for experiments.

2. " *Falsifiability test for classical nucleation theory*", Camilla Beneduce, **Diogo E. P. Pinto**, Lorenzo Rovigatti, Flavio Romano, Petr Šulc, Francesco Sciortino, and John Russo, Phys. Rev. Lett. **134**, 148201 (2025).
   Significance: We demonstrate a fundamental limitation of classical nucleation theory by showing that thermodynamically equivalent crystal polymorphs exhibit strikingly different

nucleation behaviors. Our findings challenge the capillarity approximation and highlight the crucial role of structural fluctuations in the liquid phase.

3. *" Automating Blueprints for Colloidal Quasicrystal Assembly"*, **Diogo E. P. Pinto**, Petr Šulc, Francesco Sciortino, and John Russo, ACS Nano **19**, 512 (2025).
Significance: We propose a novel automated method to self-assemble aperiodic structures in 3D, like quasi-crystals, using directional interactions with colour specificity between particle bonds.

4. *" Inverse design of self-folding 3D shells"*, **Diogo E. P. Pinto**, Nuno A. M. Araújo, Petr Šulc, and John Russo, Phys. Rev. Lett. **132**, 118201 (2023).
Significance: We introduce a novel inverse design technique which allows one to switch between competing 3D shells by controlling the self-folding pathway of 2D planar nets.

5. " *Design strategies for the self-assembly of polyhedral shells*", **Diogo E. P. Pinto**, Petr Šulc, Francesco Sciortino, and John Russo, PNAS **120**, e2219458120 (2023).
Significance: A novel inverse design technique (SAT-assembly) is used to efficiently self-assemble patchy particles into polyhedral shells, leading to increased yields and greater control over the final structure.

6. " *Two-step nucleation in a binary mixture of Patchy Particles*", Camilla Beneduce, **Diogo E. P. Pinto**, Petr Šulc, Francesco Sciortino, and John Russo, J. Chem. Phys **158**, 154502 (2023).
Significance: Simulations and mean-field theory of patchy particles show that the azeotropic point in binary mixtures enhances nucleation rates.

7. " *Cell motility in confluent tissues induced by substrate disorder*", **Diogo E. P. Pinto**, Margarida M. Telo da Gama, and Nuno A. M. Araújo, Phys. Rev. Res. **4**, 023186 (2022).
Significance: Computer simulations of epithelial monolayers highlight how substrate heterogeneity changes tissue rigidity depending on the ratio between the characteristic length scale of heterogeneity and the typical cell size.

8. " *Hierarchical structure of the energy landscape in the Voronoi model of dense tissue*", **Diogo E. P. Pinto**, Daniel M. Sussman, Margarida M. Telo da Gama, and Nuno A. M. Araújo, Phys. Rev. Res. **4**, 023187 (2022).
Significance: An epithelial monolayer model is shown to have a crossover between two rigid states with different energy landscapes, like highly compressed glasses.

9. " *The Cell Adaptation Time Sets a Minimum Length Scale for Patterned Substrates*", **Diogo E. P. Pinto**, Gonca Erdemci-Tandogan, M. Lisa Manning, and Nuno A. M. Araújo, Byophys. J. **119**, 2299 (2020).
Significance: The use of substrate patterns to compartmentalize epithelial monolayers can be compromised by the typical adaptation time scale of cells. Through simulations and a mean-field model, a maximum length scale for substrate patterning is proposed.

10. ” *Random sequential adsorption on mobile patches*”, **Diogo E. P. Pinto** and Nuno A. M. Araújo, Phys. Rev. E **98**, 012125 (2018).
Significance: The surface coverage of a substrate with mobile patches is controlled by the ratio between the particle’s adsorption flux and patch diffusion. Moreover, patch density and particle size set a crossover between uncorrelated and correlated adsorption.

11. ” *Kinetic control of the coverage of oil droplets by DNA-functionalised colloids*”, Darshana Joshi, Dylan Bargtail, Alessio Caciagli, Jerome Burelbach, Zhongyang Xing, André S. Nunes, **Diogo E. P. Pinto**, Nuno A. M. Araújo, Jasna Bruijc, and Erika Eiser, Sci. Adv. **2**, e1600881 (2016).
Significance: A protocol is proposed for the adsorption of DNA-coated colloidal particles on the surface of oil droplets leading to full control over the coverage and fully ergodic colloidal dynamics on the droplets.

Projects as Principal Investigator

- **Rheology and fracture in epithelial tissues** **2025 - 2027**

  Funded (55.197,23 €) by the Fundação para a Ciência e a Tecnologia (FCT) under reference 2024.15260.PEX, “Concurso para Projetos de Investigação de caráter Exploratório em Todos os Domínios Científicos 2024”

Organization of scientific meetings

- **Quantum, Soft, and Living Matter Workshop** **September 2026**

  Main organizer. University of Lisbon, Lisbon, Portugal.

- **CFTC Summer Internships** **July 2026**

  Main organizer. University of Lisbon, Lisbon, Portugal.

Oral communications

- **12th International Conference on Biological Physics** **June 2026**

  Genova, Italy.

- **10th International Soft Matter Conference** **May 2026**

  Goa, India.

- **10th IDMRCS** **July 2025**

  Barcelona, Spain.
- **Physics of Biological Systems Workshop** **May 2025**

  Edinburgh, United Kingdom.
- **The Physics of Collective Cell Dynamics and Morphogenesis Workshop** **April 2025**

  Oxford, United Kingdom.
- **Failure in soft materials: from yielding to fracture Workshop** **Sep 2024**

  Vienna, Austria (CECAM Flagship Workshop).
- **6th Italian Soft Matter Days** **Jul 2024**

  Florence, Italy (Awarded Soft Matter presentation prize by the Royal Society of Chemistry).
- **7th International Soft Matter Conference** **Sep 2023**

  Osaka, Japan.
- **3rd Condensed Matter Physics National Conference** **Mar 2023**

  Lisbon, Portugal.
- **CMD2020GEFES** **Aug 2020**

  Madrid, Spain (Online).
- **13º Encontro Nacional de Química-Física** **Jun 2018**

  Faro, Portugal.
- **Flowing Matter 2018** **Jan 2018**

  Lisbon, Portugal.
- **Ciência 2017** **Jul 2017**

  Lisbon, Portugal.
- **Flowing Matter 2017** **Jan 2017**

  Porto, Portugal.

Poster presentations

- **StatPhys 29** **July 2025**

  Florence, Italy.

- **Physics of Life 2025** **March 2025**
  Harrogate, United Kingdom. (Awarded poster prize by the Institute of Physics).
- **From Water To Colloidal Water** **Jun 2022**
  Rome, Italy (Awarded Soft Matter poster prize by the Royal Society of Chemistry).
- **11th Liquid Matter Conference** **Jul 2021**
  Prague, Czech Republic (Online).
- **2nd Portuguese Condensed Matter Physics Meeting** **May 2019**
  Porto, Portugal.
- **Dynamics of self-organization: from colloids to biomaterials Workshop** **Dec 2017**
  Barcelona, Spain.

Fellowships, awards and and other competitive selections

- **Porter prize winner** **2025**
  Awarded poster prize by the Institute of Physics during the Physics of Life 2025 conference.
- **Marie Skłodowska-Curie European Postdoctoral Fellowship** **2024**
  Award granted by the European Commission, with seal of excellence, for project RheoCell under call HORIZON-MSCA-2023-PF-01-01 — MSCA Postdoctoral Fellowships 2023.
- **Presentation Prize winner** **2024**
  Awarded Soft Matter presentation prize by the Royal Society of Chemistry during the 6th Italian Soft Matter Days.
- **Computer time allocation** **2023**
  Co-proposer with Prof. John Russo to the Italian SuperComputing Resource Allocation (ISCRA) at CINECA for project NANOFOLD: Design and simulation of capsid formation from self-folding nets.
- **Porter prize winner** **2022**
  Awarded Soft Matter poster prize by the Royal Society of Chemistry during the From Water To Colloidal Water conference.
- **PhD individual fellowship** **2017**
  Fellowship granted by Portuguese Foundation for Science and Technology (FCT) under no. SFRH/BD/131158/2017.

- **ERASMUS+ grant** **2015**

  Monthly salary granted by the European Commission to develop research activities at the University of Zaragosa.

Supervision and mentorship

- **Camilla Beneduce** **2022 – 2026**

  Assisting Prof. John Russo with PhD supervision: Nucleation of binary mixtures of patchy particles into diamond cubic structures.

- **Niccoló Tedeschi** **Sep 2024**

  Co-supervisor of MSc thesis project: A graph theory approach to the inverse self-assembly problem

- **Enrico DiCenso** **Mar 2024**

  Co-supervisor of MSc thesis project: Hexagonal crystal nucleation and growth using patchy particles through the SAT-assembly strategy

Teaching

- **Physics of Continuous Media** **Fall 2025 and 2026**

  Main lecturer, Faculty of Sciences, University of Lisbon, Portugal.

- **General Physics for Biochemistry** **Spring 2026**

  Assisting, Faculty of Sciences, University of Lisbon, Portugal.

- **Materials Science and Technology** **Spring 2026**

  Assisting, Faculty of Sciences, University of Lisbon, Portugal.

- **Numerical Methods** **Fall 2020, 2019, and 2017**

  Teaching assistant, Faculty of Sciences, University of Lisbon, Portugal.

- **Electromagnetism** **Spring 2019 and 2018**

  Teaching assistant, Faculty of Sciences, University of Lisbon, Portugal.

- **General Physics for Computer Science** **Spring 2017**

  Teaching assistant, Faculty of Sciences, University of Lisbon, Portugal.

- **Experimental Physics II** **Fall 2016**

  Teaching assistant, Faculty of Sciences, University of Lisbon, Portugal.

- **Optics and Electromagnetism** **Fall 2016**

  Teaching assistant, Faculty of Sciences, University of Lisbon, Portugal.

Science Communication & Outreach

CFTC summer internships **2019 and 2020**

Participated as organizer and supervisor during the CFTC summer internships for Bsc students.

Open day FCUL **2017, 2018, and 2019**

Participated in FCUL's open day, as a member of the Center for Theoretical and Computational Physics, promoting the interaction between researchers and high school students.

Jobshop FCUL **2017, 2018 and 2019**

Participated in Jobshop FCUL, as a member of the Center for Theoretical and Computational Physics, promoting the interaction between academia and industry.

Ciência 2017 **2017**

Oral presentation at the Portuguese annual science conference promoting the outreach of science to a general audience.

Professional service

- **Referee for International Journals** **Since 2017**

  Physical Review Letters (since 2026), Physics of Fluids (since 2025), Soft Matter (since 2024), The Journal of Chemical Physics (since 2022), Physical Review E (since 2021), International Journal of Modern Physics C (since 2017).

- **Junior Assessor MPhys projects** **2025**

  University of Oxford, Oxford, United Kingdom.